

Shankar Mallesh

MSc Data Science & AI · Data Engineering · AI Developer

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Right to Work in the UK — Student Visa (no sponsorship required for internship roles)

PROFILE

MSc Data Science & AI student at Aston University with 2+ years of professional experience designing and delivering production data pipelines, SQL schemas, and AI-integrated data systems. Proven ability to own data engineering work end-to-end — from schema design and pipeline automation through to monitoring and troubleshooting in live environments. Passionate about turning complex data into reliable infrastructure that drives real business decisions.

EDUCATION

MSc Data Science and Artificial Intelligence

Aston University, Birmingham, UK · *April 2026 – April 2027 (Expected)*

- Key modules: Data Pipeline Architecture, Machine Learning, NLP, Large Language Models, Knowledge Representation, Distributed Systems & Algorithms
- Applying schema design, retrieval architecture and pipeline automation directly to live production projects alongside academic study

Bachelor of Computer Applications (BCA)

Bangalore University, India · First Class · *2019 – 2022*

- Core studies: Database Systems, SQL & Data Modelling, Algorithms & Data Structures, Software Engineering, Operating Systems

CERTIFICATIONS

Python Developer Certification

[freecodecamp.org](https://www.freecodecamp.org) · *May 2026*

- Completed ~300 hours of structured curriculum covering Python fundamentals, data structures, algorithms, and applied projects — directly reinforcing production experience with Pandas, NumPy, and Scikit-learn

Verify: freecodecamp.org/certification/fcc-c6094a8d.../python-v9

TECHNICAL SKILLS

Data & Databases: SQL, PostgreSQL, Supabase, Firebase (NoSQL), data schema design, data warehouse concepts

Data Engineering: Pipeline design & automation, ETL processes, structured & unstructured data processing, data quality monitoring

Languages: Python (Pandas, NumPy, Scikit-learn), JavaScript, TypeScript, Dart

AI & Analytics: LLM Integration, RAG, NLP, Agentic Workflows, Generative AI, Prompt Engineering

Tools & Practices: Git/GitHub, REST APIs, Clean Architecture, Agile/Scrum, Technical Documentation, Code Review

EXPERIENCE

Full-Stack AI Developer — CircleSave

Freelance · Remote, USA · May 2025 – Present

- Designed and implemented production PostgreSQL schemas for a digital savings circle (ROSCA) platform, modelling complex relational data for member management, contribution tracking, payout scheduling, and full audit logging
- Built JWT-based authentication with OTP email verification and role-based access control (Owner / Member), enforcing backend validation of all critical financial actions to prevent unauthorised access and duplicate contributions
- Developed automated financial workflows covering contribution tracking, payment confirmation, payout scheduling, and cycle completion validation — ensuring fairness and transparency in multi-member savings rounds
- Implemented a real-time notification system for join requests, approvals, and cycle updates; owned ongoing pipeline monitoring and defect resolution to maintain system reliability in production
- Built with Next.js, TypeScript, Supabase (PostgreSQL), and Nodemailer; collaborated directly with the client to translate business requirements into backend-driven data models and automated workflows

Junior Software Developer

ARK Power Solutions Pvt. Ltd., India · Onsite · Aug 2022 – May 2024

- Supported data migration activities across production systems, contributing to data integrity validation and troubleshooting during complex migrations
- Developed cross-platform applications using Flutter, building scalable features from a shared codebase and reducing delivery time by ~30% through clean architecture
- Collaborated within Agile engineering teams on debugging, code reviews and documentation — developing strong practices around data handling and software quality

PROJECTS

Aplypse — Template Marketplace for Builders

Solo Project · Actively in development · Aug 2024 – Present

- Conceived and built a full-stack template marketplace where every listing ships as a complete, scalable product concept — each entry includes a product brief, a futuristic UI, and clean source code ready to clone and build on
- Implemented the core catalog experience: browsable template cards with live previews, concept briefs explaining the idea and its potential, category filtering (SaaS, fintech, health, productivity), and one-click download / git clone for instant source code access
- Gated downloads behind a lightweight free account layer to maintain quality control; applied a strict three-bar template philosophy (concept must be real and sellable, design must be ahead of its time, code must be clean enough to build on)
- Shipped v1 solo — lean and fast — with a deliberate roadmap: v2 targets an AI news & tools directory, v3 introduces community submissions and image generation competitions

INTERESTS

Geopolitics and international affairs — particularly the intersection of technology, policy and global power. Avid reader across non-fiction, history and technology. Keen interest in responsible AI, data ethics and the societal implications of large-scale data systems.

REFERENCES

Keerthan Pagadala — Associate Director, ARK Power Solutions Pvt. Ltd., India

Professional and academic references available upon request.